

Your LAST Name _____

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Your Collaborators

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Policy Statement

- We encourage you to collaborate, but only in a group of up to *five* current EE 120 students.
- Be sure to write the names of your collaborators at the appropriate location on the next page. Each collaborator must submit a distinct, self-prepared solution document containing original contributions to the collaborative effort.
- To submit your work, you have several options:
 - (a) Print out this homework document and write your solutions on the printout. Then upload a *PDF file* of the scan.
 - (b) Write your solutions on this PDF file using a tablet device. Then upload a *PDF file*.
 - (c) Use your own sheets of paper, but demarcate clearly the space for each problem, part, and subpart. Then upload a *PDF file*.
- Please write neatly and legibly, because *if we can't read it, we can't evaluate it*. Remember that, even though the problem sets are self-graded, we will evaluate your work whenever we check your self grades.
- Unless we explicitly state otherwise, you may expect full credit *only if* you explain your work succinctly, but clearly and convincingly.
- If you are asked to provide a “sketch,” it refers to a well-labeled *hand-drawn* sketch—not a plot generated by a computing device.
- On occasion, a problem set contains one or more problems designated as “optional.” Your solutions to such problems will NOT be evaluated. Nevertheless, you’re responsible for learning the subject matter within their scope.
- If you have a confirmed disability that precludes you from complying fully with these instructions or with any other parameter associated with this problem set, please alert us immediately about reasonable accommodations afforded to you by the DSP Office on campus.

Overview

This problem set explores

- DT State Space
- The Dirac delta

HW5.1 (A Jolly Famous Sequence of Numbers)

The input x and the output y of a causal, discrete-time LTI system H are governed by the second-order linear constant-coefficient difference equation

$$y[n] = y[n-1] + y[n-2] + x[n],$$

where the system is assumed to be initially at rest—that is, $y[n] = 0$ for all $n < 0$.

- (a) Determine $h[n]$, the impulse response values of the system H . What famous sequence of numbers does the impulse response h trace? Is the system BIBO stable? Provide a succinct, yet clear and convincing explanation.

Plug in $x[n] = \delta[n]$.

$$\begin{aligned} H \text{ causal} &\implies h[n] = 0, \quad \forall n < 0 \\ h[0] &= h[-1] + h[-2] + \delta[0] = 1 \\ h[1] &= h[0] + h[-1] + \underbrace{\delta[1]}_0 = 1 \\ h[2] &= h[1] + h[0] = 2 \\ h[3] &= h[2] + h[1] = 3 \\ h[4] &= h[3] + h[2] = 5 \\ &\vdots \end{aligned}$$

The impulse response is the Fibonacci sequence.

This system is not BIBO stable, as the impulse response keeps on growing with each sample and is therefore not finitely summable.

- (b) Draw a delay-adder-gain (DAG) block diagram implementation of the system.

Because of the $y[n-2]$ term, at least two delay blocks are needed to implement the system. There are a couple DAGs that can be drawn for this problem, with two of them shown below.

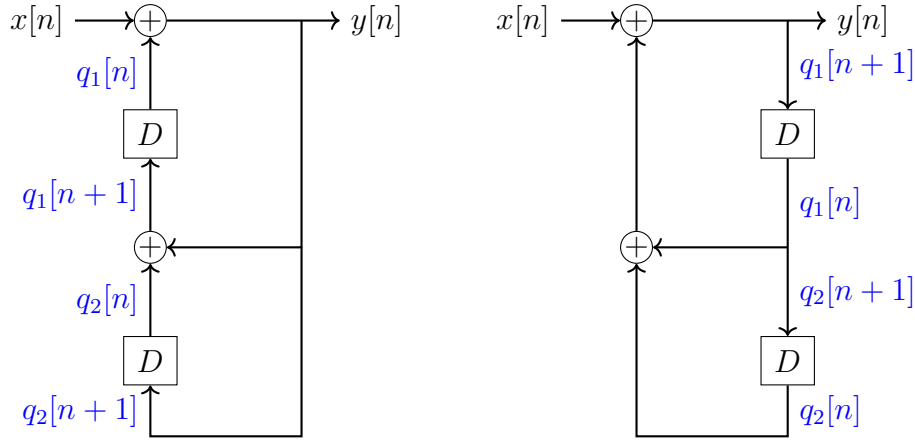


Figure 1: Two possible delay-adder-gain block diagrams for 4.1(b)

- (c) On the DAG block diagram implementation of the previous part, label the appropriate state variables $q_1[n]$ and $q_2[n]$. Then proceed to write the corresponding state-space equations

$$\mathbf{q}[n+1] = \mathbf{A} \mathbf{q}[n] + \mathbf{B} x[n] \quad \text{State-Evolution Equation}$$

and

$$y[n] = \mathbf{C} \mathbf{q}[n] + \mathbf{D} x[n], \quad \text{Output Equation}$$

where $\mathbf{q}[n] = [q_1[n] \ q_2[n]]^\top$ is the state vector at time n . You must, of course, determine the matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ completely.

We denote where our state variables are on the diagrams in Figure 1. For the diagram on the left,

$$\begin{aligned} q_1[n] &\triangleq y[n-1] + y[n-2] \\ q_2[n] &\triangleq y[n-1] \\ \begin{bmatrix} q_1[n+1] \\ q_2[n+1] \end{bmatrix} &= \begin{bmatrix} y[n] + q_2[n] \\ y[n] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} q_1[n] \\ q_2[n] \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\mathbf{B}} x[n] \\ y[n] &= q_1[n] + x[n] = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{C}} \mathbf{q}[n] + \underbrace{1}_{\mathbf{D}} x[n], \quad \mathbf{D} \triangleq 1 \end{aligned}$$

For the diagram on the right,

$$\begin{aligned} q_i[n] &\triangleq y[n-i], \quad i = 1, 2 \\ \begin{bmatrix} q_1[n+1] \\ q_2[n+1] \end{bmatrix} &= \begin{bmatrix} y[n] \\ y[n-1] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} q_1[n] \\ q_2[n] \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\mathbf{B}} x[n] \\ y[n] &= \underbrace{\begin{bmatrix} 1 & 1 \end{bmatrix}}_{\mathbf{C}} \mathbf{q}[n] + x[n] \end{aligned}$$

Note that if we swap our definitions for $q_1[n]$ and $q_2[n]$ for either DAG block diagram, we will get $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, which is a valid setup but will not match the format for the modes in part (d).

- (d) Determine the modes $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ of the system. For consistency, show that the modes are of the form $\left(\lambda_1, \mathbf{v}_1 = \begin{bmatrix} 1 \\ -\lambda_2 \end{bmatrix}\right)$ and $\left(\lambda_2, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -\lambda_1 \end{bmatrix}\right)$, where λ_1 and λ_2 are the eigenvalues that you must determine. Also, explain why the eigenvectors *must* be mutually orthogonal.

Since an eigenvalue λ and eigenvector $\vec{v} \neq \vec{0}$ of $\mathbf{A}$ satisfy $\mathbf{A}\vec{v} = \lambda\vec{v}$, $\lambda\mathbf{I} - \mathbf{A}$ has a nontrivial nullspace and $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$.

$$\det \begin{bmatrix} \lambda - 1 & -1 \\ -1 & \lambda \end{bmatrix} = \lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} : (\lambda_1 \mathbf{I} - \mathbf{A}) \vec{v}_1 = \begin{bmatrix} \frac{-1+\sqrt{5}}{2} & -1 \\ -1 & \frac{1+\sqrt{5}}{2} \end{bmatrix} \vec{v}_1 = \vec{0}$$

$$\vec{v}_1 = \begin{bmatrix} 1 \\ \frac{-1+\sqrt{5}}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ -\lambda_2 \end{bmatrix}$$

$$\lambda_2 = \frac{1 - \sqrt{5}}{2} : (\lambda_2 \mathbf{I} - \mathbf{A}) \vec{v}_2 = \begin{bmatrix} \frac{-1-\sqrt{5}}{2} & -1 \\ -1 & \frac{1-\sqrt{5}}{2} \end{bmatrix} \vec{v}_2 = \vec{0}$$

$$\vec{v}_2 = \begin{bmatrix} 1 \\ \frac{-1-\sqrt{5}}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ -\lambda_1 \end{bmatrix}$$

The eigenvectors *must* be orthogonal because $\mathbf{A}$ is a symmetric matrix. By the Spectral Theorem, we can decompose a symmetric matrix: $\mathbf{A} = \mathbf{V}\mathbf{D}\mathbf{V}^{-1}$. $\mathbf{V}$ is an orthonormal matrix containing the eigenvectors of $\mathbf{A}$, so the eigenvectors of $\mathbf{A}$ are orthogonal.

- (e) Is the system *internally stable*? Provide a succinct, yet clear and convincing explanation.

The system is not internally stable because the mode associated with λ_1 “blows up” as $n \rightarrow \infty$. Since $\mathbf{A}$ is diagonalizable, we can look at the state-evolution

equation under a change of basis.

$$\begin{aligned}
\mathbf{A} \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} &= \underbrace{\begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix}}_{\triangleq \mathbf{V}} \underbrace{\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}}_{\triangleq \mathbf{D}} \\
\vec{q}[n+1] &= \mathbf{V} \mathbf{D} \mathbf{V}^{-1} \vec{q}[n] \\
\underbrace{\mathbf{V}^{-1} \vec{q}[n+1]}_{\triangleq \vec{z}[n+1]} &= \mathbf{D} \underbrace{\mathbf{V}^{-1} \vec{q}[n]}_{\triangleq \vec{z}[n]} \\
\vec{z}[n+1] &= \mathbf{D} \vec{z}[n] \\
\begin{bmatrix} z_1[n+1] \\ z_2[n+1] \end{bmatrix} &= \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} z_1[n] \\ z_2[n] \end{bmatrix} \\
&= \begin{bmatrix} \lambda_1 z_1[n] \\ \lambda_2 z_2[n] \end{bmatrix} \\
z_i[n+1] &= \lambda_i z_i[n], \quad i = 1, 2 \\
z_i[n] &= \lambda_i^{n-1} z_i[1]
\end{aligned}$$

Since $|\lambda_1| = \frac{1+\sqrt{5}}{2} \approx 1.618 > 1$ and $z_1[1] \neq 0$, $\lim_{n \rightarrow \infty} |z_1[n]| = \infty$.

HW5.2 (Dirac Delta Properties)

(a) **(Sampling Property)** Simplify, and provide a well-labeled sketch of, each of the following expressions involving the Dirac delta:

(i) $(t^2 + 1)\delta(t)$

We recall that $\delta(t) = 0 \forall t \neq 0$. Therefore, $\forall t \neq 0, (t^2 + 1)\delta(t) = 0$ and for $t = 0, (t^2 + 1)\delta(t) = \delta(t)$. So,

$$(t^2 + 1)\delta(t) = (0^2 + 1)\delta(t) = \delta(t).$$

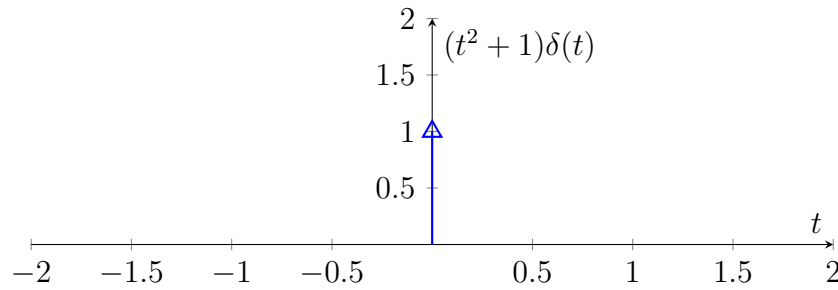


Figure 2: Plot of $(t^2 + 1)\delta(t)$ for 4.2(a)(i)

As the Dirac Delta has an infinite height but finite area, we will indicate the area values in plots as their “height” as if it is a Kronecker delta. Note that this is an important distinction to make.

(ii) $e^{\alpha t} \delta(t - 1)$, where $\alpha \in \mathbb{C}$

For this problem, note that $\delta(t - 1) = 0 \forall t \neq 1$. Hence, $e^{\alpha t} \delta(t - 1) = 0 \forall t \neq 1$, so that

$$e^{\alpha t} \delta(t - 1) = e^{\alpha} \delta(t - 1).$$

$\alpha \in \mathbb{C}$, so $e^{\alpha} = e^{\Re(\alpha)} + i e^{\Im(\alpha)}$ is a complex number with magnitude $e^{\Re(\alpha)}$ and phase $\Im(\alpha)$. Let us plot the magnitude and phase of $e^{\alpha} \delta(t - 1)$:

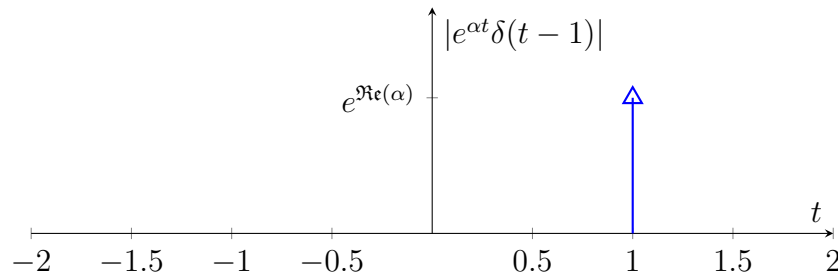


Figure 3: Plot of $|e^{\alpha t} \delta(t - 1)|$ for 4.2(a)(ii)

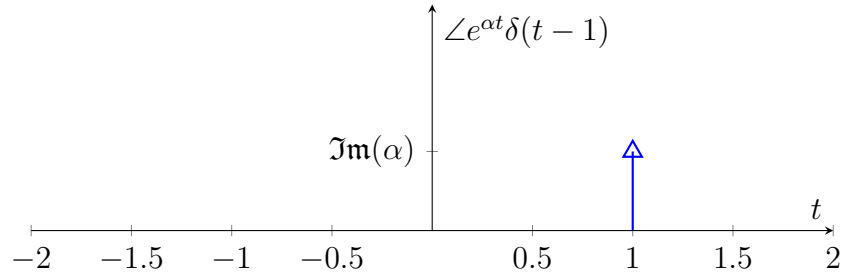


Figure 4: Plot of $\angle e^{\alpha t} \delta(t-1)$ for 4.2(a)(ii)

(iii) $\cos\left(\pi t + \frac{\pi}{2}\right) \delta\left(t - \frac{1}{2}\right)$

Again, we simply plug in the value such that $\delta(t - 1/2) \neq 0$, so that

$$\begin{aligned} \cos\left(\pi t + \frac{\pi}{2}\right) \delta\left(t - \frac{1}{2}\right) &= \cos\left(\pi \frac{1}{2} + \frac{\pi}{2}\right) \delta\left(t - \frac{1}{2}\right) \\ &= \cos(\pi) \delta\left(t - \frac{1}{2}\right) \\ &= -\delta\left(t - \frac{1}{2}\right). \end{aligned}$$

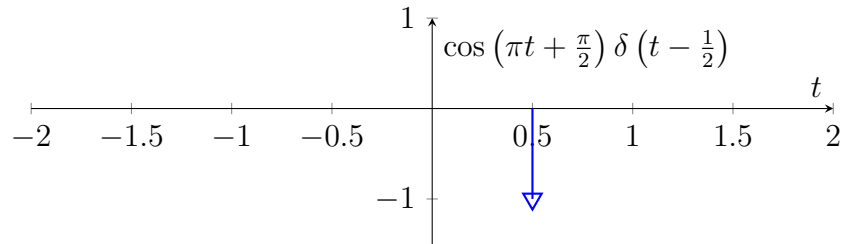
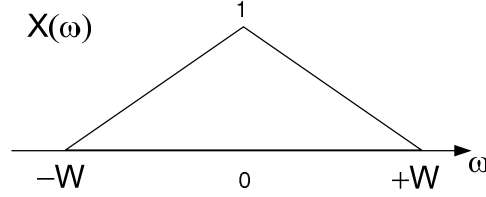


Figure 5: Plot of $\cos\left(\pi t + \frac{\pi}{2}\right) \delta\left(t - \frac{1}{2}\right)$ for 4.2(a)(iii)

(iv) $X(\omega) P(\omega)$, where P is the impulse train

$$P(\omega) = \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{kW}{2}\right),$$

with $W > 0$, and $X(\omega)$ is as shown below:



The function $P(\omega) = \sum_{k=-\infty}^{\infty} \delta\left(\omega - k\frac{W}{2}\right)$ has a Dirac delta impulse at every $\frac{W}{2}$ time steps. So that,

$$X(\omega)P(\omega) = \sum_{k=-\infty}^{\infty} X\left(k\frac{W}{2}\right) \delta\left(\omega - k\frac{W}{2}\right).$$

The function $X(\omega)$ is only non-zero for $|\omega| < W$, so that $\forall |k| > 1$, $X\left(k\frac{W}{2}\right) = 0$. Furthermore, for the given function, $X\left(\frac{W}{2}\right) = X\left(-\frac{W}{2}\right) = \frac{1}{2}$ and $X(0) = 1$, so

$$X(\omega)P(\omega) = \delta(\omega) + \frac{1}{2} \left(\delta\left(\omega + \frac{W}{2}\right) + \delta\left(\omega - \frac{W}{2}\right) \right).$$

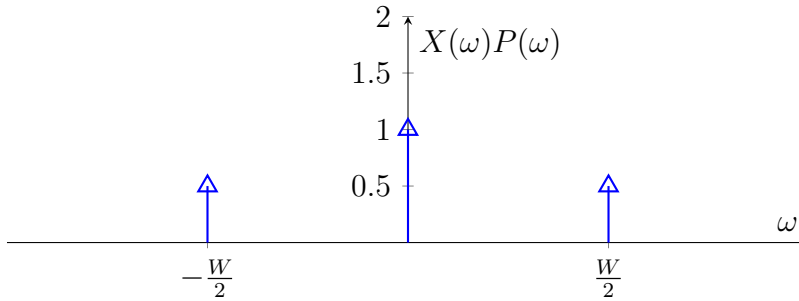


Figure 6: Plot of $X(\omega)P(\omega)$ for 4.2(a)(iv)

(b) **(Sifting Property)** Evaluate each of the following expressions involving the Dirac delta:

(i) $\int_{-\infty}^{+\infty} e^{\alpha t} \delta(t-1) dt$, where $\alpha \in \mathbb{C}$

$$\int_{-\infty}^{\infty} e^{\alpha t} \delta(t-1) dt = e^{\alpha}$$

$$(ii) \int_{0.49}^{0.51} \sin(\pi t) \delta\left(t - \frac{1}{2}\right) dt$$

$$\int_{0.49}^{0.51} \sin(\pi t) \delta\left(t - \frac{1}{2}\right) dt = \sin(\pi/2) = 1$$

$$(iii) \int_{-\infty}^{+\infty} X(\omega) \delta\left(\omega - \frac{2W}{3}\right) d\omega, \text{ where } X(\omega) \text{ is as shown in part (a)(iv) above.}$$

$$\begin{aligned} \int_{-\infty}^{\infty} X(\omega) \delta\left(\omega - \frac{2W}{3}\right) d\omega &= X\left(\frac{2W}{3}\right) \\ &= \frac{1}{3} \end{aligned}$$

$$(iv) \int_{-\infty}^{+\infty} e^{-|t|} v(t) dt, \text{ where } v \text{ is the impulse train}$$

$$v(t) = \sum_{k=-\infty}^{+\infty} \delta(t - k).$$

For this problem we let $v(t) = \sum_{k=-\infty}^{\infty} \delta(t - k)$, so that

$$\int_{-\infty}^{\infty} e^{-|t|} v(t) dt = \int_{-\infty}^{\infty} e^{-|t|} \sum_{k=-\infty}^{\infty} \delta(t - k) dt.$$

Bringing the exponential into the summation, yields

$$\int_{-\infty}^{\infty} e^{-|t|} \sum_{k=-\infty}^{\infty} \delta(t - k) dt = \int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} e^{-|t|} \delta(t - k) dt.$$

From the above problems, we know that $e^{-|t|} \delta(t - k) = e^{-|k|} \delta(t - k)$. Thus,

$$\int_{-\infty}^{\infty} e^{-|t|} \sum_{k=-\infty}^{\infty} \delta(t - k) dt = \int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} e^{-|k|} \delta(t - k) dt.$$

Now, by the definition of Dirac deltas, we have that

$$\int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} e^{-|k|} \delta(t - k) dt = \sum_{k=-\infty}^{\infty} e^{-|k|} = 1 + 2 \sum_{k=1}^{\infty} e^{-|k|}.$$

Keeping in mind the formula for a geometric series (with geometric ratio between -1 and 1), we can rewrite and solve for the summation as the

following.

$$\begin{aligned}\sum_{k=-\infty}^{\infty} e^{-|k|} &= \sum_{k=-\infty}^0 e^k + \sum_{k=0}^{\infty} e^{-k} - e^0 \\ &= 2 \left(\frac{1}{1 - e^{-1}} \right) - 1 \\ &= \frac{2 - 1 + e^{-1}}{1 - e^{-1}} = \frac{e + 1}{e - 1}.\end{aligned}$$

HW5.3 (Dilation, Contraction, and Time-Reversal Properties)

Show that scaling the argument of the Dirac delta affects its strength (area) in the following manner:

$$\delta(at) = \frac{1}{|a|} \delta(t), \quad \forall a \in \mathbb{R}. \quad (1)$$

Equation (1)—which denotes an operational equivalence, not a point-wise equality—is sometimes referred to as the *time-scaling* property, even though the argument of the Dirac delta need not be “time.”

To verify the time-scaling property, place each side of Equation (1) inside an integral with a test function g that is continuous at $t = 0$, and verify that both sides produce the same result. That is, show that

$$\int_{-\infty}^{+\infty} \delta(at) g(t) dt = \int_{-\infty}^{+\infty} \frac{1}{|a|} \delta(t) g(t) dt. \quad (2)$$

Hint: Use the change of variable $\tau = at$ in the left-hand side of Equation (2).

Show that the Dirac delta *behaves like* an even function, namely, $\delta(-t) = \delta(t)$. Note that this is true even if the function that the Dirac delta idealizes is not even.

We consider two different cases: $a > 0$ and $a < 0$. We use the substitution of variables $\tau = at$.

For $a > 0$:

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(at) g(t) dt &= \int_{-\infty}^{\infty} \delta(\tau) g(\tau/a) \frac{1}{a} d\tau \\ &= \frac{1}{a} \int_{-\infty}^{\infty} \delta(\tau) g(\tau/a) d\tau \\ &= \frac{1}{a} g(0) \\ &= \frac{1}{a} \int_{-\infty}^{\infty} \delta(t) g(t) dt. \end{aligned}$$

For $a < 0$:

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(at) g(t) dt &= \int_{\infty}^{-\infty} \delta(\tau) g(\tau/a) \frac{1}{a} d\tau \\ &= \frac{-1}{a} \int_{-\infty}^{\infty} \delta(\tau) g(\tau/a) d\tau \\ &= \frac{-1}{a} g(0) \\ &= \frac{-1}{a} \int_{-\infty}^{\infty} \delta(t) g(t) dt. \end{aligned}$$

Note that for the case of $a > 0$, we have $\frac{1}{a}$, but for the case of $a < 0$, we have $\frac{-1}{a}$. This occurs in the negative a case since the substitution of variables $\tau = at$ flips the

bounds of the integral. In both cases, we have the scaling factor to be a positive value. Therefore, we can summarize the equation for both cases as

$$\int_{-\infty}^{\infty} \delta(at)g(t)dt = \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(t)g(t)dt.$$

To show that $\delta(t) = \delta(-t)$, plug in $a = -1$ into $\delta(at) = \frac{1}{|a|}\delta(t)$ to get:

$$\delta(-t) = \frac{1}{|-1|}\delta(t) = \delta(t).$$